

- 5 The I – V characteristic of an electrical component X is shown in Figure 5.1.

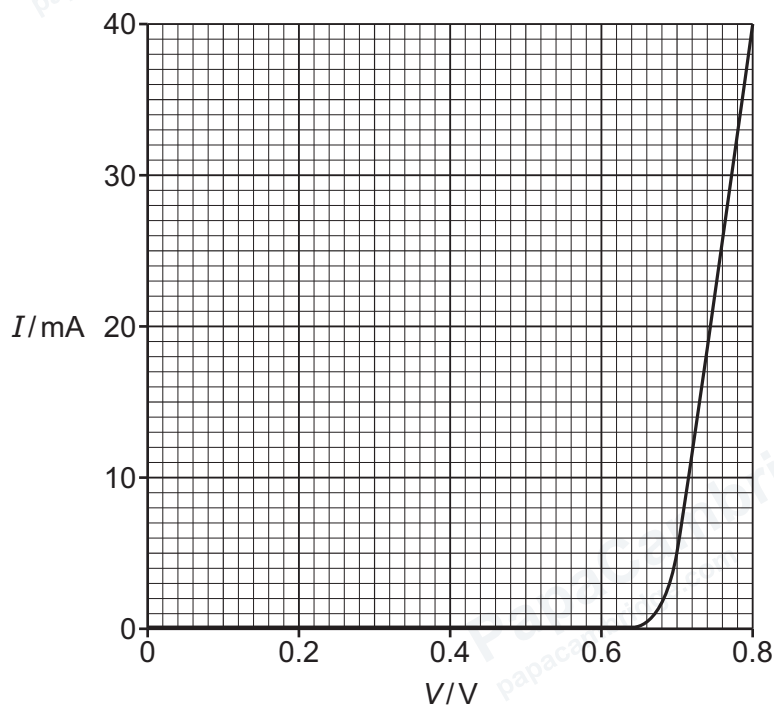


Figure 5.1

- (a) State the name of electrical component X.

..... [1]

- (b) A metallic conductor has a constant resistance of $40\,\Omega$. On Figure 5.1, sketch the I – V characteristic of this conductor between $V = 0$ and $V = 0.80\,\text{V}$.

[2]

- (c) Component X is connected in a circuit with a resistor of resistance $20\ \Omega$ as shown in Figure 5.2.

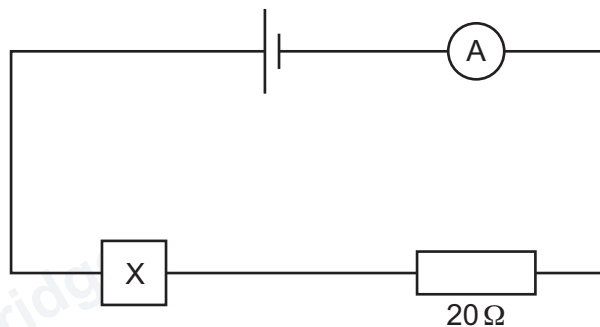


Figure 5.2

The current in the ammeter is 5.0 mA and the cell has negligible internal resistance.

Show that the electromotive force (e.m.f.) of the cell is 0.80 V .

[3]

- (d) The terminals of the cell in 5(c) are now reversed without making any other changes to the circuit.

State the reading on the ammeter.

reading on ammeter =A [1]

- (e) The resistor of resistance $20\ \Omega$ and component X are now connected in parallel to the cell in 5(c) in the circuit shown in Figure 5.3.

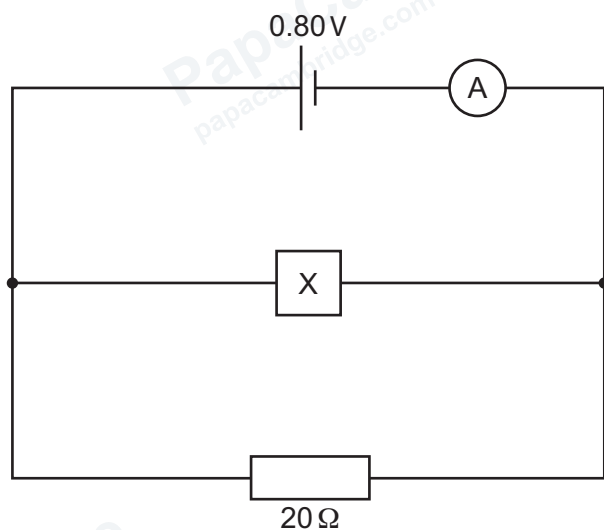


Figure 5.3

The current in component X is **not** negligible.

Use Figure 5.1 to determine the current in the ammeter.

current =A [3]

[Total: 10]